



Influence of plant composition and water use strategies on green roof stormwater retention

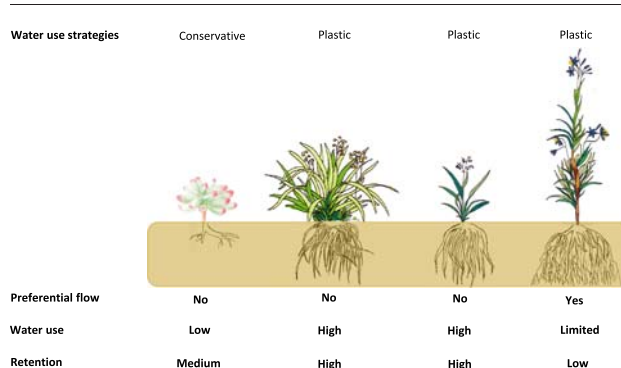
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HIGHLIGHTS

- We evaluated how plant water use strategies influenced ET and stormwater retention.
- ET and retention were greatest in plants with high water use and drought tolerance.
- Plant roots reduced retention and soil water content due to preferential flow.
- Preferential flow overwhelmed the influence of water use strategies.

GRAPHICAL ABSTRACT



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ABSTRACT

Green roofs are increasingly being considered a promising engineered ecosystem for reducing stormwater runoff. Plants are a critical component of green roofs and it has been suggested that plants with high water use after rainfall, but which are also drought tolerant, can improve rainfall retention on green roofs. However, there is little evidence to show how plants with different water use strategies will affect green roof retention performance, either in monocultures or in mixed plantings. This study tested how monocultures and a mixture of herbaceous species (*Dianella admixta*, *Lomandra longifolia* and *Styphandra glauca*) affected rainfall retention on green roofs. These species were chosen based on their water use strategies and compared with a commonly used succulent species (*Sedum pachyphyllum*) with conservative water use. We measured retention performance for 67 rainfall events, quantifying all components of the water balance. We also compared growth for species in monocultures and mixtures. We found that monocultures of *L. longifolia* had the greatest stormwater retention and ET. Although *S. glauca* has a similar water use strategy to *D. admixta*, it had the lowest stormwater retention and ET. In both the mixture and as a monoculture, *S. glauca* created preferential flow pathways, resulting in lower substrate water contents which reduced ET and therefore rainfall retention. This species also dominated performance of the mixture, such that the mixture had lower ET and retention than all monocultures (except *S. glauca*). We suggest that root traits and their interaction with substrates should be considered alongside water use strategies for rainfall retention on green roofs.

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1. Introduction

Impervious surfaces in urban areas increase the frequency, volume and peak flow of stormwater runoff compared with natural areas

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which degrades urban aquatic ecosystems (Walsh et al., 2005). As roofs comprise a large proportion of impermeable surfaces, green roofs are being used as a tool to manage urban runoff (Li and Babcock, 2014; Elliott et al., 2016). Although they provide other ecosystem services, such as air purification, noise reduction and urban cooling, reducing stormwater runoff is often considered the major benefit provided by green roofs and is one of the main reasons they are legislated or incentivised in cities worldwide (Oberndorfer et al., 2007; Carter and Fowler, 2008).

Green roofs reduce stormwater runoff by storing water during rainfall events, delaying peak runoff (attenuation through the substrate) and removing water from the substrate to the atmosphere through evapotranspiration (ET) (Stovin et al., 2013). Following rainfall, a portion of the detained water drains from the roof and the amount of rainfall corresponding to substrate storage capacity is retained (Bengtsson et al., 2005). This retained water will then either evaporate or be transpired by plants which dries out the substrate and regenerates retention capacity before the next rainfall event (Berretta et al., 2014; Poë et al., 2015). The rate at which water stored in the substrate is used by plants is an important determinant of how a green roof responds to the next rainfall event (Villarreal and Bengtsson, 2005). The most commonly planted species on green roofs are succulents in the genus *Sedum*. These plants are ideal for survival in green roof systems, because their shallow root system and conservative water use strategy means that they can withstand frequent periods of drought on green roofs when the shallow substrates dry out (VanWoert et al., 2005; Getter and Rowe, 2009). However, these characteristics also mean that they may not sufficiently dry out substrates via transpiration between rain events to provide significant stormwater retention (Farrell et al., 2013). To optimise rainfall retention and plant survival on green roofs, plants need to have high transpiration when water is available after rainfall, but also be able to survive dry substrates between rainfall events by limiting their transpiration (Wolf and Lundholm, 2008; Farrell et al., 2013). These water use strategies have been found in rock outcrop plants evaluated for green roofs, but their potential for reducing stormwater runoff has not yet been evaluated (Farrell et al., 2013).

It has been suggested that a diverse mixture of plant functional groups will improve the function and resilience of green roofs (Lundholm et al., 2010). Studies in both natural and engineered ecosystems have found positive relationships between plant diversity and the provision of ecosystem services, including: maintaining hydrologic cycles, cleaning air and water and the storing/cycling of nutrients (Christensen et al., 1996; Quijas et al., 2010; Johnson et al., 2016). Two principal mechanisms are thought to drive the positive relationships between species diversity and ecosystem function: (1) transgressive overyielding, where mixtures of plant species produce more biomass than monocultures due to complementarity in resource utilization (Schmid et al., 2008; Prieto et al., 2015); and (2) selection effect, where there is a greater possibility of including the best-performing species in a mixture which enhances resource use efficiency (Chapin III et al., 1997). Species which have similar effects on ecosystem processes (water and nutrient dynamics, trophic interactions, or disturbance regime), but differ in response to environmental conditions, can also provide stability (resistance and resilience) under changing environmental conditions (Chapin III et al., 2000). Consequently, if one species is lost from a diverse ecosystem, there is the potential for other functionally similar species to compensate, making the ecosystem more resilient to environmental changes (Ehrlich and Ehrlich, 1981).

Applying biodiversity and ecosystem theory to improve the performance of green roofs has been gaining traction, with several studies evaluating how plant diversity affects rainfall retention (Dunnett et al., 2008; Lundholm et al., 2010). However, most of these studies have evaluated diversity in terms of mixtures of life-forms or species richness, rather than by mixing species with known functionality, such as the water use strategy. While these studies hypothesised that mixtures would out-perform monocultures in terms of retention, the results

have not been straight forward. For example, Lundholm et al. (2010) reported that while mixtures of grasses and succulents had greater retention than monocultures of either life-form, their retention was less than bare substrate. The effects of life-form diversity are likely complicated by differences in biomass and rainfall interception. For example, Dunnett et al. (2008) found there was no significant improvement in retention with life-form diversity, but that the highest retention was from tall plants with high root biomass. This is consistent with Franzaring et al. (2016) which showed that, compared with monocultures of legumes, herbs, small shrubs and succulents, mixtures of these life-forms retained less water than monocultures of grasses. While species richness seems to be positively related to productivity, it may not relate to rainfall retention (Nagase and Dunnett, 2012; Johnson et al., 2016).

These results are consistent with ecological research that shows that not all metrics of biodiversity are related to improved nutrient and water uptake, biomass production and transpiration rate (Díaz and Cabido, 2001; Lavorel and Garnier, 2002). It has been suggested that functional diversity and functional trait diversity may be the most important drivers for specific ecosystem services (Petchey and Gaston, 2002; Cadotte et al., 2011). Consequently, green roof experiments assessing the role of diversity on the basis of life-forms or species richness may not have included the necessary functional or physiological traits to improve water retention. This leads us to consider whether bio-diverse green roofs designed on the basis of functional diversity or functional trait diversity could achieve the trade-off between hydrological performance and plant survival. We therefore suggest that green roofs planted with mixtures of species with appropriate water use strategies will improve rainfall retention performance without compromising plant survival (Farrell et al., 2013).

In this study, we tested how mixtures and monocultures of species chosen on the basis of plant water use strategies affect rainfall retention on green roofs, relative to a commonly used succulent species. We evaluated three species (*Lomandra longifolia*, *Dianella admixta* and *Stypanandra glauca*) known to have water use plasticity, i.e. they have high evapotranspiration when water is available and reduced water use under drought conditions to avoid drought stress (Farrell et al., 2013). We predicted that: (1) green roof modules planted with monocultures of species with water use plasticity would retain more water via greater evapotranspiration than modules planted with the succulent species with conservative water use and (2) mixtures would retain more rainfall due to complementarity in aboveground biomass and water use, resulting in greater evapotranspiration.

2. Materials and methods

2.1. Experimental design

We used a randomized block design with six 'vegetation' treatments and three replicates: (1) bare substrate (as a control), (2) monoculture of *Sedum pachyphyllum* (as a reference for common green roof plantings), (3) monoculture of *Lomandra longifolia*, (4) monoculture of *Dianella admixta*, (5) monoculture of *Stypanandra glauca* and (6) mixture of *Lomandra longifolia*, *Dianella admixta* and *Stypanandra glauca*. These four species were chosen based on two types of water use strategies, (1) conservative water use: *S. pachyphyllum* and (2) plastic water use: *L. longifolia*, *D. admixta* and *S. glauca* (Table 1) (Farrell et al., 2013).

The experiment was located at the Burnley Campus of The University of Melbourne, Australia (−37.828472, 145.020883). Melbourne has monthly average maximum and minimum temperatures of 19.9 and 10.2 °C (Melbourne Regional Office; site 086071; Australian Bureau of Meteorology). We used 18 modules (6 treatments with 3 replicates per treatment) with dimensions 1.15 × 1.15 m, placed in an open-ended rainfall exclusion shelter (i.e., a poly tunnel; 28 m long × 2.6 m high). Modules were constructed with drainage, geotextile and substrate layers to simulate a typical green roof structure. Modules were filled by weight with 105 kg of a scoria-based substrate to a depth of

Table 1

Water use strategies of plant species used in this experiment, as determined from a glass-house study under well-watered (WW) and water deficit (WD) conditions (Farrell et al., 2013).

Family	Species	Water use (g H ₂ O d ⁻¹)		Water use strategy
		WW	WD	
Liliaceae	<i>Dianella admixta</i>	71.0	17.5	Plastic
Lomandraceae	<i>Lomandra filiformis</i>	39.9	11.3	Plastic
Phormiaceae	<i>Stypandra glauca</i>	85.1	21.9	Plastic
Crassulaceae	<i>Sedum pachyphyllum</i>	4.3	5.2	Conservative

100 mm. The substrate was tested for air-filled porosity (13.8%), bulk density (1.26 g cm⁻³) and water holding capacity (45.9%) before planting. These properties satisfied the FLL guidelines which are commonly used to assess the suitability of green roof substrates (FLL, 2008; Farrell et al., 2013). Plants were planted as tubestock in June 2012 at a density of 18 plants/module and grown for 36 months before the experiment started (Fig. 1). This meant that the plants were representative of an established green roof with well-developed root systems and canopy cover. Aside from rainfall due to the rainout shelter, modules were under natural climatic conditions including temperature, wind and radiation.

2.2. Rainfall simulation

We ran a rainfall simulation for 468 days, from 14 July 2015 to 25 October 2016. To apply a 'typical' year rainfall based on Melbourne's recent climate observations, we sourced long-term daily rainfall data for Melbourne (Melbourne Regional Office; site 086071; Australian Bureau of Meteorology) for 50 years (1964–2014) from SILO (Jeffrey et al., 2001). We calculated the 50th percentile rainfall depth for each season, then combined the four seasons to make a "typical" year. In total, 92 rainfall events were applied over the 468 days of the experiment. The number of dry days between two subsequent rainfall events ranged between zero and 20 days.

Rainfall was applied with an overhead irrigation system with 11 rows of 121 sprinklers positioned 1.32 m above the substrate surface, applying 0.76 mm min⁻¹ to each module. We measured applied

rainfall, retention and runoff from simulated rain events to compare hydrological performance of different vegetation treatments.

2.3. Runoff collection and calculating total rainfall retention

Runoff was collected from each module during each simulated rainfall event in PVC pipes constructed to form a U-shape (Fig. 1). One side of the 'U' was connected to the outlet of each module and the other side was fitted with a 0.5 m Odyssey capacitance water level sensor (Datafow Systems Ltd., Christchurch, NZ). Sensors recorded the change in water level depth at 30 second intervals. Runoff volumes were calculated as the total change in water level depth multiplied by the area of the pipe. Rainfall retention was calculated by rainfall minus runoff for each event and total rainfall retention was calculated as the sum of retention for all events. We also calculated the average runoff delay for all events which produced runoff, i.e., the time-delay between the start of the rainfall event and the initial measurement of runoff. The average time to initiation of runoff was calculated to show how vegetation treatments influenced rainfall movement through substrate. Runoff data was collected for 67 of the 92 events applied during the experiment.

2.4. Evapotranspiration and soil water content

To compare evapotranspiration among treatments, we measured the weights of modules using a load cell (HBM RSCC s-type; with 50 g resolution and maximum capacity of 200 kg) attached to a gantry crane before and after each rainfall event (Fig. 1). ET between rainfall events was determined from the difference in module weight 24 h (to allow for runoff collection) after each event and immediately before the next event. Cumulative ET was determined as the total amount of water lost from each module during the experiment.

We also calculated the soil water content (SWC) before and after each rainfall event from module weights, according to:

$$SWC_t = \frac{(M_{tot} - M_{fp} - M_{ds} - M_{ms})}{A_m} \quad (1)$$

where SWC_t is the soil water content at time t, M_{tot} is the total weight of total module, M_{fp} is the final fresh weight of plants, M_{ds} is the weight of dry substrate, M_{ms} is the weight of module structure and A_m is the area

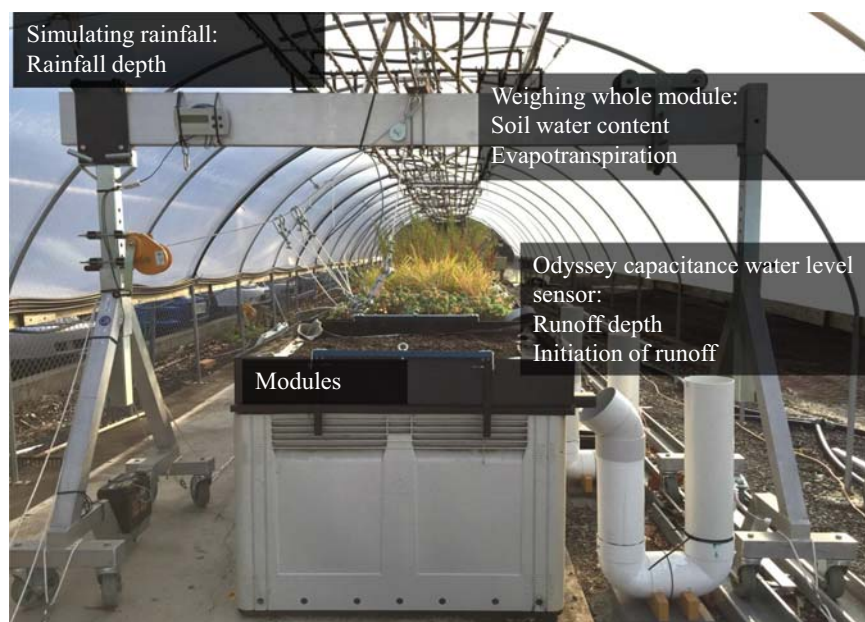


Fig. 1. Experimental apparatus showing planted and bare green roof modules under the rain shelter and overhead irrigation for application of simulated rainfall events. The U-shape pipes contain water level sensors to monitor runoff. The gantry crane was used to weigh modules before and after each event.

of the module. We calculated an average soil water content for the duration of the experiment.

2.5. Plant biomass at the end of the experiment

At the end of the experiment, we randomly selected and harvested three similar, representative plants of each species from each module. There were no noticeable differences in condition among the selected plants. Shoot biomass was removed and weighed (fresh weight), then dried in an oven at 105 °C for 72 h (dry weight). Root biomass from each of the selected plants was removed using a 150 mm diameter PVC pipe core, hammered into the full depth of the substrate and centred on the removed shoot. Roots were washed from the substrate and patted dry using paper towel before determining fresh weights, then dry weights (after drying in a 70 °C oven for 72 h). From these measurements, we calculated the average shoot and root biomass per plant for each module. For the total module root biomass, we multiplied the average root biomass of the three harvested plants by the number of plants per module. To determine the total module shoot biomass, we harvested all remaining shoot biomass and added this to the shoot biomass of the three individual plants harvested.

2.6. Substrate properties at the end of the experiment

We also measured whether differences in water retention among treatments were related to changes in substrate properties over time. Three substrate core samples (74 mm diameter and 100 mm deep) were randomly extracted from each module and tested for water holding capacity (WHC), air-filled porosity (AFP) and bulk density at field capacity according to the Australian Standard for Potting Mixes AS 3743-2003 (Standards Australia, 2003).

2.7. Statistical analyses

The effect of planting treatments on retention, cumulative ET, SWC, initiation of runoff and (shoot and root) biomass were analysed using one-way ANOVA. Tukey's HSD was used to indicate significant differences among treatments ($P < 0.05$). Relationships between all variables, regardless of treatment, were also determined from correlation analysis. Data were transformed where necessary (all data shown are non-transformed). All analyses were undertaken with R version 3.3.1 (R Core Team, 2016), including packages 'dplyr', 'tidyr', 'ggplot2', 'lubridate', 'cowplot' and 'agricolae'.

3. Results

3.1. Total rainfall retention

Rainfall retention differed significantly among treatments (Fig. 2A). Modules planted with *L. longifolia* had the highest retention, with 16% greater retention than *S. glauca* and 10% greater retention than *S. pachyphyllum*. Interestingly, *S. glauca*, not *S. pachyphyllum* which has a conservative water use strategy, had the lowest retention. Modules planted with *S. glauca* and the mixture of species, also had lower rainfall retention than the bare treatment. Modules planted with mixtures retained less rainfall than monocultures of *L. longifolia*.

3.2. Cumulative evapotranspiration

Modules with *L. longifolia*, which retained more rainfall than any other treatment, also had the highest cumulative ET (Fig. 2B). Patterns in cumulative ET were very similar to those observed for retention, where *L. longifolia* had significantly higher cumulative ET than *S. glauca*. The difference between the lowest (*S. glauca* and mixture) and the highest (*L. longifolia*) cumulative ET was 33%. Cumulative ET

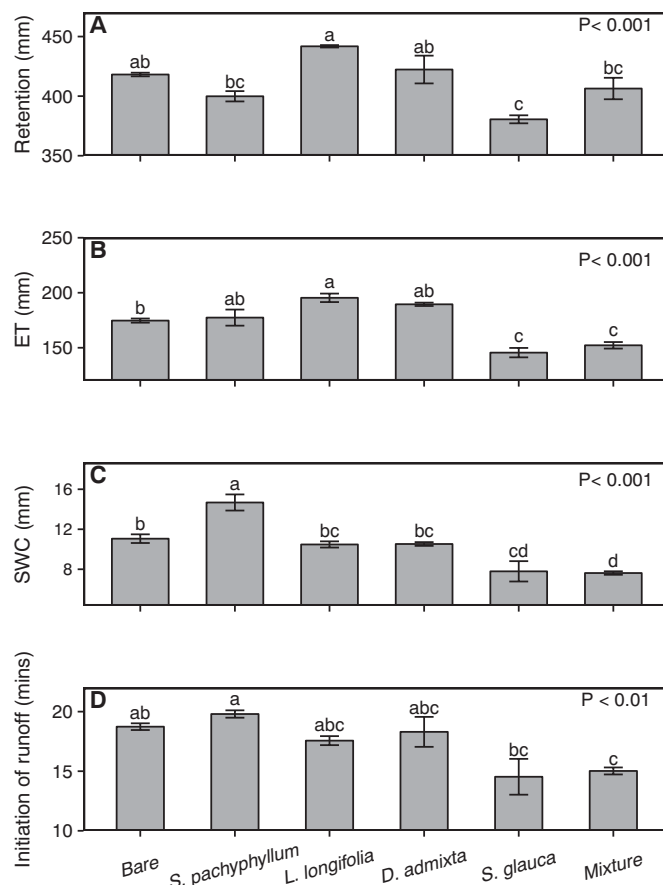


Fig. 2. Mean retention (A), evapotranspiration (ET; B), soil water content (SWC; C) and time to initiation of runoff (D) for all treatments. Different letters denote significant differences among means (one-way ANOVA, $n = 3$; all $P < 0.001$) and bars represent mean standard error.

of the mixture was significantly lower than that of all other treatments except *S. glauca*.

3.3. Average soil water content

Average SWC results also reflected the results for retention and ET. Average SWCs of the mixture and the monoculture planting of *S. glauca* were the lowest, while the *S. pachyphyllum* treatment had the highest average SWC (Fig. 2C).

3.4. Substrate properties at the end of the experiment

There were no differences in substrate properties, including: air-filled porosity (AFP), water holding capacity (WHC) and bulk density among treatments at the end of the experiment (Table 2).

Table 2

Mean ($\pm$ SE; $n = 3$) substrate air filled porosity (AFP), water holding capacity (WHC) and bulk density at field capacity of the green roof modules at the end of the experiment.

Treatment	AFP (%)	WHC (%)	Bulk density (g cm^{-3})
Bare	12.77 (± 1.70)	36.50 (± 2.17)	0.91 (± 0.07)
<i>S. pachyphyllum</i>	13.18 (± 1.71)	36.62 (± 0.78)	0.90 (± 0.04)
<i>L. longifolia</i>	14.65 (± 2.19)	37.02 (± 1.85)	0.85 (± 0.02)
<i>D. admixta</i>	15.05 (± 0.91)	38.46 (± 1.55)	0.86 (± 0.03)
<i>S. glauca</i>	14.07 (± 1.73)	39.19 (± 0.58)	0.83 (± 0.03)
Mixture	16.12 (± 1.29)	40.85 (± 0.12)	0.76 (± 0.01)
P-value	0.722	0.249	0.185

P-values indicate no significant differences between treatments for a one-way ANOVA.

3.5. Average initiation of runoff

Although there were no differences in substrate properties at the end of the experiment, there were differences in the average time to initiation of runoff after rainfall was applied. Modules planted with monocultures of *S. glauca* and the mixture had runoff occurring 27% and 25% faster than modules with the slowest time to initiation of runoff (*S. pachyphyllum*; Fig. 2D).

3.6. Influence of cumulative ET, average SWC, substrate WHC, initiation of runoff and biomass on retention

Correlation analysis showed that retention was positively correlated with cumulative ET and the initiation of runoff (Table 3). The initiation of runoff was strongly and positively correlated with cumulative ET and average SWC, but negatively correlated with the substrate WHC at the end of the experiment. There were no significant relationships between shoot or root biomass or the root/shoot ratio and retention, cumulative ET, average SWC or initiation of runoff.

3.7. Biomass

At the module level, *S. pachyphyllum* and *L. longifolia* had the highest shoot biomass, followed by *S. glauca* and the mixture, while *D. admixta* had the lowest shoot biomass (Fig. 3A). *D. admixta* had the highest root biomass, whereas *S. pachyphyllum* had the lowest root biomass (Fig. 3C). *L. longifolia*, *S. glauca* and the mixture had similar root biomass (Fig. 3C).

When grown together in the mixture, *L. longifolia* and *S. glauca* had similar shoot biomass, and were significantly larger than *D. admixta* (Fig. 3B). Not all plants showed different biomass on a per plant basis when grown in a monoculture compared with the mixture (Fig. 3B and D). *L. longifolia* had similar shoot as well as root biomass when grown as a monoculture and in the mixture (Fig. 3B and D). In contrast, both shoot and root biomass of *D. admixta* were significantly lower when growing in the mixture compared with when growing as a monoculture (Fig. 3B and D). *S. glauca* behaved very differently; its shoot biomass was 50% higher in the mixture compared with the monoculture (Fig. 3B). However, it had similar root biomass for both monocultures and mixtures (Fig. 3D). Of the three species in the mixture, *S. glauca* had the largest shoot biomass and root biomass (Fig. 3B and D).

4. Discussion

We aimed to test how mixtures and monocultures of species chosen on the basis of plant water use strategies would affect rainfall retention on green roofs, relative to a commonly used succulent species. We predicted that (1) green roof modules planted with species with water use plasticity would retain more water via greater ET than those planted with the succulent *S. pachyphyllum* which has a conservative water

use strategy and (2) mixtures would retain more rainfall due to complementarity in aboveground biomass and water use, resulting in greater ET.

4.1. Retention was not always higher in species with plastic water use strategies

In this study, there was on average 411.5 mm rainwater retained by green roof modules and this accounted for 73% of the applied rainfall. The range of rainfall retention reported from other similar green roof studies (across more than one year) is between 5% and 80% (Mentens et al., 2006; Voyde et al., 2010; Stovin et al., 2012; Carson et al., 2013; Li and Babcock, 2014; Locatelli et al., 2014; Cipolla et al., 2016; Elliott et al., 2016; Sims et al., 2016; Szota et al., 2017). Compared with these studies, the climate in Melbourne is relatively warm and dry, and this probably explains why our average retention is at the higher end of this range. For example, Sims et al. (2016) showed that retention performance is not only influenced by rainfall amount but also by ET, with the greatest retention where high ET rates reduce antecedent soil moisture. The high retention in our study is also consistent with a previous study on rainfall retention in the southern Australian city of Adelaide, where a 100 mm deep green roof retained 74% of annual rainfall due to the influence of prolonged dry period (Razzaghamanesh et al., 2014).

Species which had previously being defined as being able to adjust their water use according to availability (plastic water use strategy) did not always show greater ET and rainfall retention than the conservative water user, *S. pachyphyllum*. This is in contrast to our previous study which suggested that green roofs planted with species able to modify their water use according to availability would have higher ET and rainfall retention (Farrell et al., 2013). We used the same species as Farrell et al. (2013), but our results may have differed as we applied a more realistic watering regime which exposed plants to discrete and episodic watering events, separated by varying periods of water deficit. Under these conditions, plants may not have sufficiently recovered when watered after periods of water deficit, limiting their water use compared with continuous well-watered conditions. The conservative water using *S. pachyphyllum* had similar ET and retention to the species with plastic water use strategies. Similar results have been demonstrated by Wolf and Lundholm (2008), who found *Sedum* species had greater retention than native wetland species and species differences only emerged when water availability was low. These results suggest that green roof plant selection for rainfall retention should not be done on the basis of water use strategies alone.

Interestingly, *S. glauca* and the species mixture had lower ET and retention compared with the *S. pachyphyllum* monoculture. Modules planted with *S. glauca* (monocultures and mixtures) also had the two lowest average SWC during the experiment. This suggests that the lower ET and retention of modules containing *S. glauca* was due to a reduction in the ability of the substrate to hold water (i.e., lower WHC). However, there was no difference among modules in the WHC of the

Table 3

Pearson correlation matrix for: retention, cumulative evapotranspiration (ET), average soil water content (SWC), water holding capacity (WHC), initiation of runoff, and shoot and root biomass variables.

	ET (mm)	SWC (mm)	WHC (%)	Initiation of runoff (mins)	Shoot biomass (g module ⁻¹)	Root biomass (g module ⁻¹)
Retention (mm)	0.756***	0.156	−0.308	0.482*	0.123	0.297
	ET	0.599**	−0.359	0.651***	0.108	0.176
		SWC	−0.446	0.728***	0.380	−0.475
			WHC	−0.627**	−0.327	0.273
				Initiation of runoff	0.228	−0.220
					Shoot biomass	−0.784***
						Root biomass

n = 18 (ET, SWC, WHC and initiation of runoff); n = 15 (shoot biomass and root biomass).

* $P \leq 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

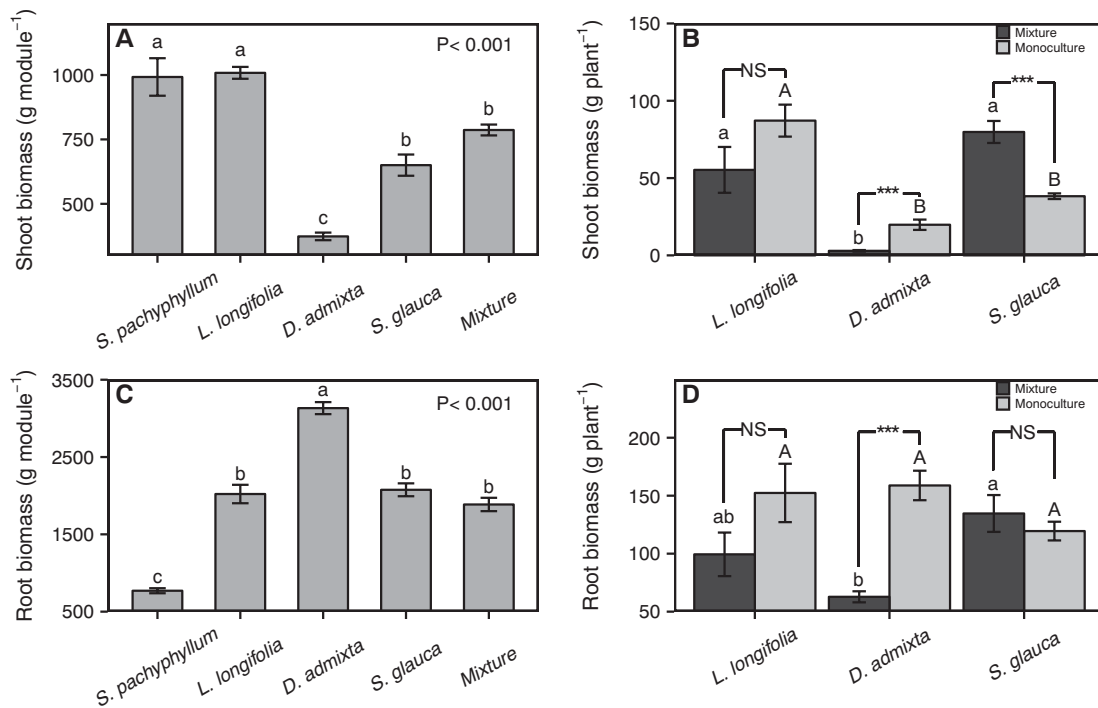


Fig. 3. Mean shoot and root biomass for all treatments at the end of the experiment, expressed on a per module basis (A and C) and on a per plant basis (B and D). For A and C, different letters denote significant difference between treatment means (one-way ANOVA, $n = 3$). For B and D, different letters show significant differences among species, within either monocultures (uppercase) or mixtures (lowercase) from two separate one-way ANOVAs ($n = 9$). Asterisks denote significant differences (one-way ANOVA) within each species, depending on whether they were growing in a mixture or as a monoculture and 'NS' means not significant at the $P = 0.05$ level. Bars represent mean standard error.

substrate at the end of the experiment. Modules planted with *S. glauca* had a shorter time to the initiation of runoff following rainfall. Because *S. glauca* plants did not change the WHC of substrates, we hypothesise that their root systems may have reduced the average SWC (and therefore ET and retention) by creating preferential flow pathways through the substrate. Root systems including decayed and living roots can increase water flux in vertical and lateral (downslope) directions via preferential flow paths by creating macropores and well-developed channel networks (Johnson and Lehmann, 2006; Carpenter and Hallam, 2009). As root biomass was not related to SWC, ET, retention or initiation of runoff, root distribution and structure (e.g., length, angle, density and diameter) are likely to be more important than simple root biomass for water retention in green roof substrates (Ghestem et al., 2011). Preferential flow pathways are generally considered positively in biofiltration systems as they maintain the infiltration capacity of filter media and reduce the adverse impact of sediment clogging over time (Hatt et al., 2009; Le Coustumer et al., 2012; Wardynski and Hunt, 2012). However, in shallow green roof substrates, preferential flow pathways reduce retention capacity. Therefore, when selecting species to maximize the stormwater benefits of green roofs we recommend considering root traits and their interaction with substrates as well as water use strategies or leaf traits related to ET.

4.2. Mixtures did not always retain more water than monocultures

We hypothesised that mixtures would retain more water than monocultures, due to greater biomass and therefore ET for mixed species plantings. This is because ecological theory suggests that higher diversity can increase resource retention through complementarity and facilitation in resource utilization (Loreau, 1998; Petchey, 2003; Cardinale et al., 2007; Schmid et al., 2008; Prieto et al., 2015); an increased possibility of including the best performing species (Loreau, 1998; Hector et al., 2002) or through greater biomass and productivity resulting in greater resource use (Schmid et al., 2008). However, this was not the case in our experiment. While, the shoot biomass of the

mixture was greater than monocultures of *D. admixta* and *S. glauca*, retention in the mixture was lower than all monocultures except *S. glauca*. In the mixture, *S. glauca* was the dominant species with the greatest shoot and root biomass among species planted in mixture, indicating that it would contribute the most to the performance of mixture. As *S. glauca* had the lowest SWC, ET and retention, likely due to its root system causing preferential flow of water from the modules before the substrate reached field capacity, it reduced the performance of the mixture relative to the monocultures of all species except itself. Therefore, the 'selection effect' of diversity in our experiment resulted in lower water resource retention.

Furthermore, biomass in our experiment was not a good predictor of ET and retention, as although *D. admixta* had the lowest biomass, both in monoculture and mixture, it did not have the lowest retention and ET. This suggests that even though diversity or species richness may positively influence biomass, there was no relationship between biomass and retention (Johnson et al., 2016). This is consistent with Lundholm et al. (2010), who also found no relationship between biomass and water use, despite greater biomass in mixtures. Therefore, for rainfall retention of mixtures on green roofs, increased biomass may be less important than selecting species based on their function and functional traits.

These findings indicate that the competitive success of *S. glauca* may be due to its greater drought tolerance achieved through storage allocation or through interference competition, but not that of water uptake necessarily. This has also been found in natural ecosystems where in some cases key species manipulated the overall function of the plant community but did not contribute to overall growth or productivity (Hooper et al., 2005). *S. glauca* was the key species in our experiment playing an important role in the biomass productivity of the mixture, but also likely introducing preferential flow, resulting in the lowest water use and retention. This is consistent with suggestions that the sampling effect of diversity is not always straightforward: with species enhancing some ecosystem services but decreasing others (Lyons and Schwartz, 2001; Tilman et al., 2001).

5. Conclusions

Contrary to our hypotheses, green roof modules planted with species with water use plasticity did not always retain more water than the conservative water using succulent, *S. pachyphyllum*. Also, the mixtures of plant species did not retain more water than monocultures. Although the mixed species treatment had higher biomass and productivity, the performance of the mixture was adversely affected by the key species, *S. glauca*. The root systems of *S. glauca* likely created preferential flow pathways through the substrate, limiting substrate water contents, evapotranspiration and rainfall retention. To optimise rainfall retention and plant survival, species selection for green roofs should therefore not only consider water use strategies or species diversity. Root traits should also be considered when selecting green roof plants, due to the importance of preferential flows overriding the benefits of plastic plant water use strategies.

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